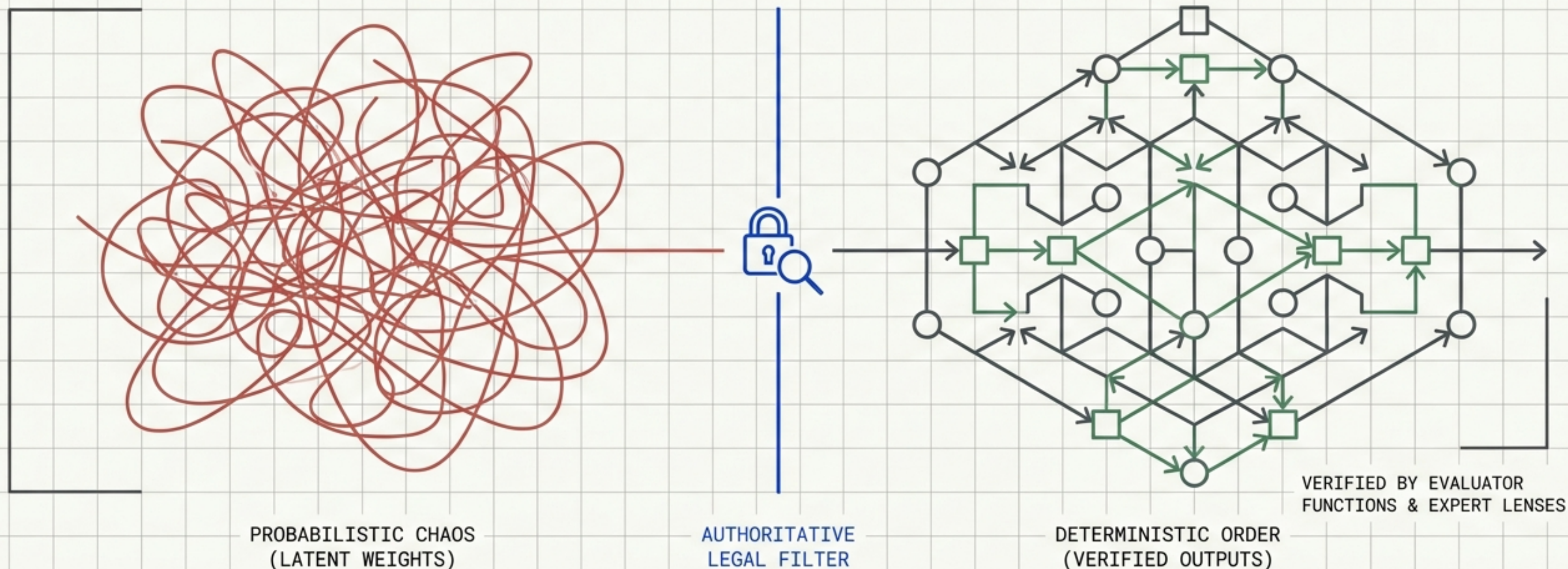


Wiring Legal Truth into Production AI

A blueprint for deterministic agents using authoritative regulatory knowledge, evaluator functions, and expert lenses.



THE FAILURE MODE IS CONFIDENCE, NOT SILENCE.

Standard LLMs hallucinate compliance because they rely on internet training data, not authoritative sources.
An agent that gives wrong compliance advice with the same fluency as correct advice is a legal liability.

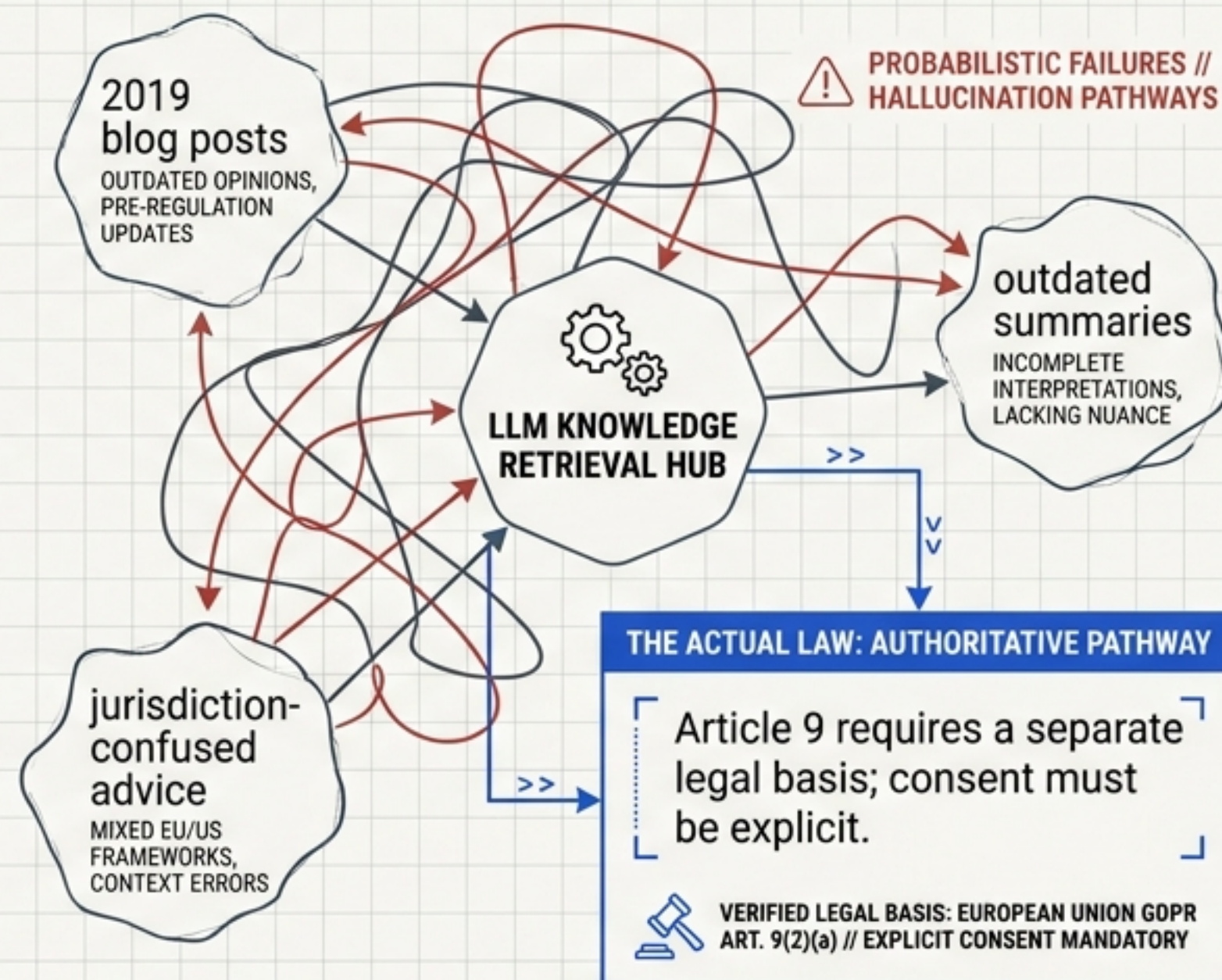
USER INTERFACE SIMULATION

Health data processing
is compliant. You have user
consent under GDPR
Article 9.

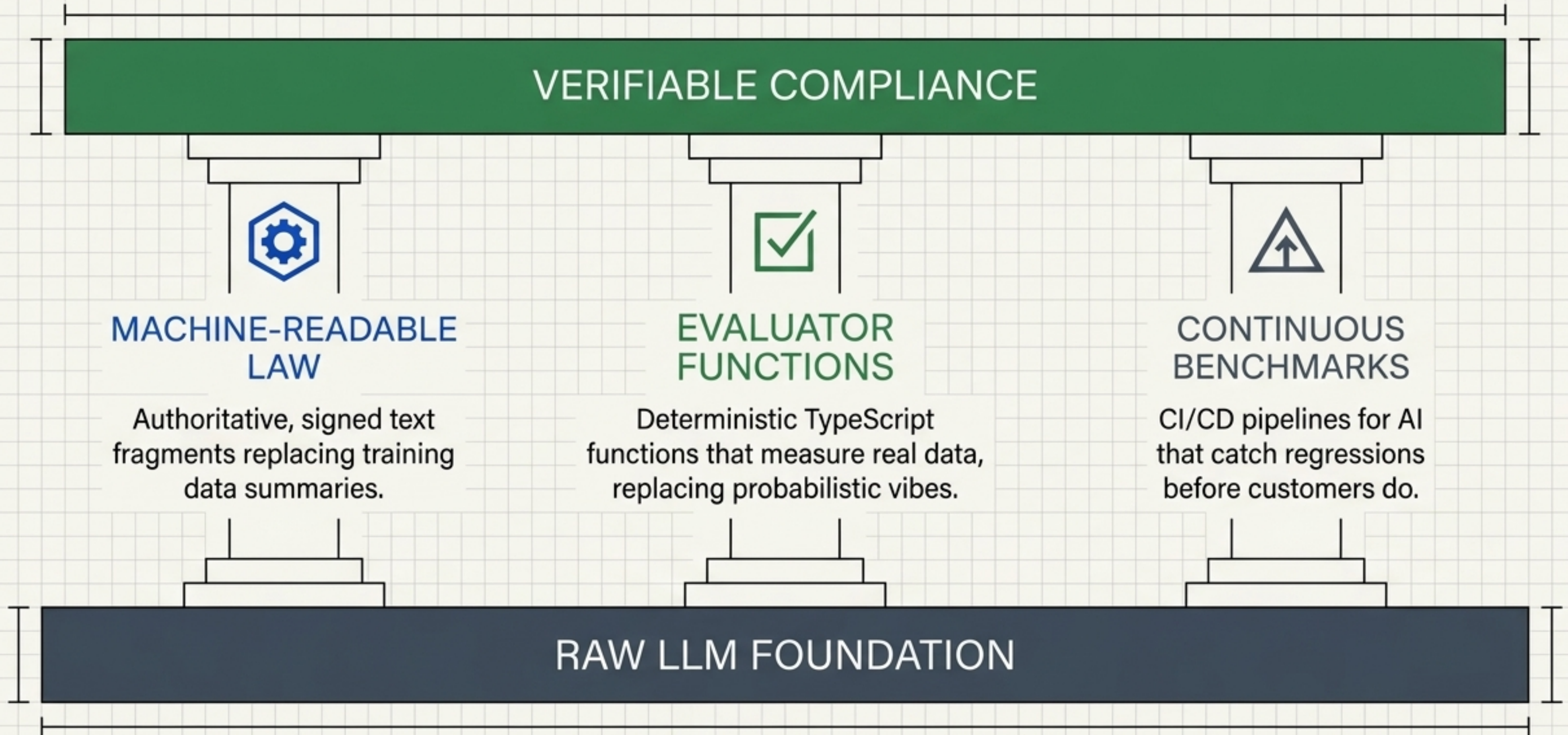
FLUENT BUT WRONG

SYSTEM OUTPUT: 99.8% CONFIDENCE SCORE (FLUENCY METRIC)

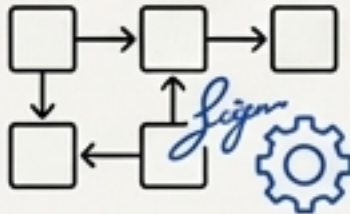
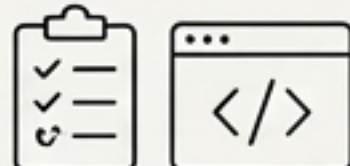
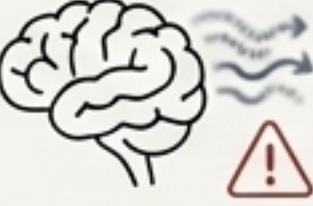
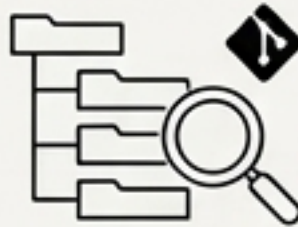
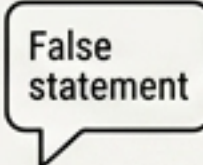
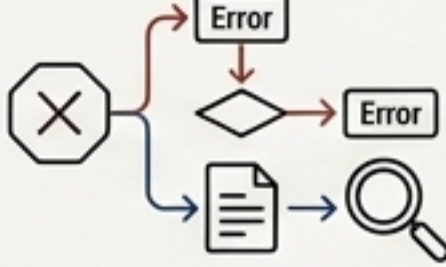
UNDERLYING KNOWLEDGE ARCHITECTURE: X-RAY VIEW



THE ARCHITECTURE OF TRUTH



PROBABILISTIC VIBES VS. DETERMINISTIC VERIFICATION

STANDARD AI SETUP	COMPLIANT AGENT ARCHITECTURE
Internet Training Data & Latent Weights PROBABILISTIC & OUTDATED 	Cryptographically Signed KCP Fragments ✓ AUTHORITATIVE & TRACEABLE 
Human Spot-Checks & Vibes INCONSISTENT & SUBJECTIVE 	178+ TypeScript Evaluator Suites AUTOMATED & REPRODUCIBLE 
LLM Approximation FLUID & IMPRECISE 	Version-Controlled YAML Expert Lenses ✓ STRUCTURED & AUDITABLE 
 Plausible, Confident Hallucination HIGH RISK OF ERROR 	 Deterministic Rejection & Error Tracing ✓ ZERO-TOLERANCE & SAFE 

THE KNOWLEDGE CONTEXT PROTOCOL (KCP)

You cannot load 88 pages of GDPR into context. The Knowledge Context Protocol (KCP)

isolates precise, article-level fragments. Cryptographic signatures guarantee the agent reads the exact, un-tampered law—not a hallucinated variation.

Trigger Fields

Routes queries to exact articles (e.g., gdpr-art-006 for legal basis).

```
---
kcp_version: "1.0"
doc_id: "eu-gdpr-2018"
fragment_id: "gdpr-art-006"
type: "law_fragment"

routing:
  trigger_fields:
    - "gdpr-art-006"
    - "legal_basis"
    - "consent_conditions"

content:
  intent_string: "Defines the legal basis for processing personal
    data, including consent, contract, and legitimate interests.
    Load when query seeks justification for data processing."
  source_text: >
    1. Processing shall be lawful only if and to the extent
    that at least one of the following applies: (a) the data
    subject has given consent... (b) processing is necessary
    for the performance of a contract...

trust:
  content_integrity:
    algorithm: "Ed25519"
    public_key: "ed25519:public:..."
    signature: "0x3a7b9e1f...c8d4a2e6"
```

Intent String

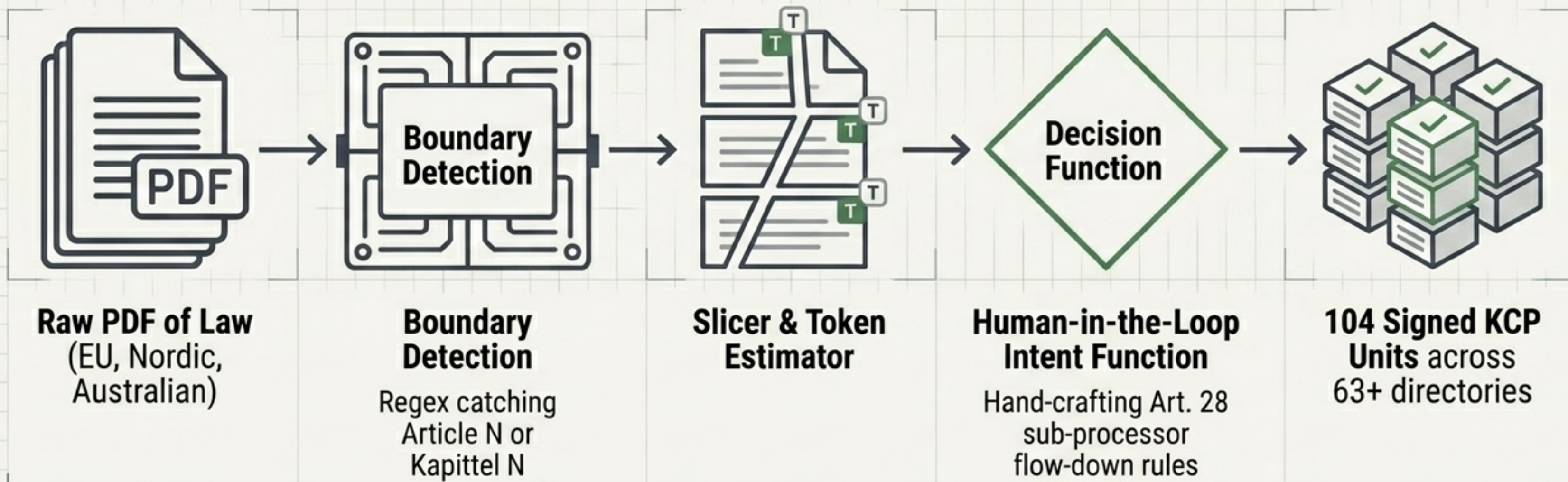
Plain-language rules telling the agent WHEN to load this chunk.

Trust Block

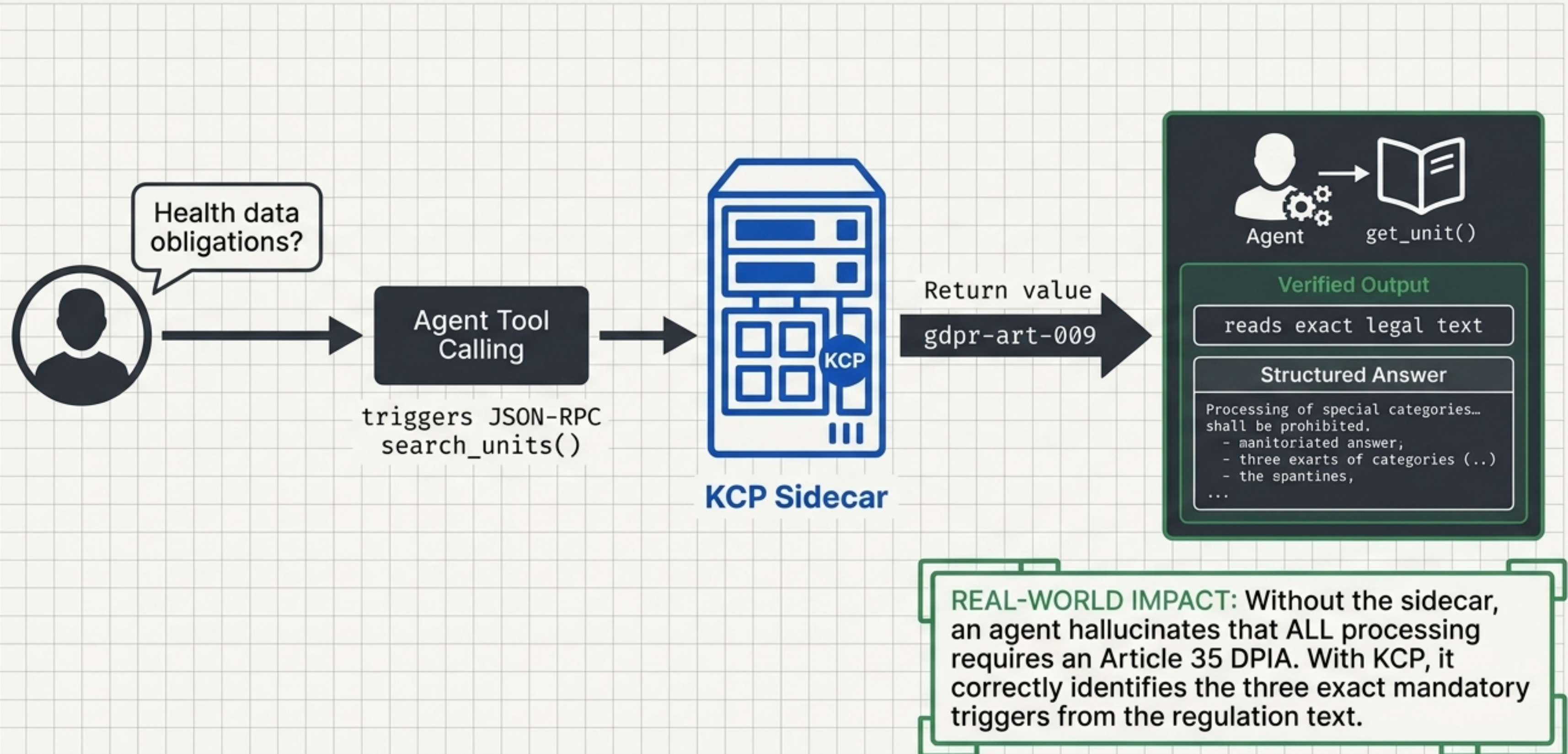
trust.content_integrity
showing a detached Ed25519
cryptographic signature.

THE AUTOMATED EXTRACTION PIPELINE

Scaling across jurisdictions requires structured boundary detection and human-in-the-loop intent mapping. Precision routing dictates whether an agent loads the right article or hallucinates a summary of twenty.

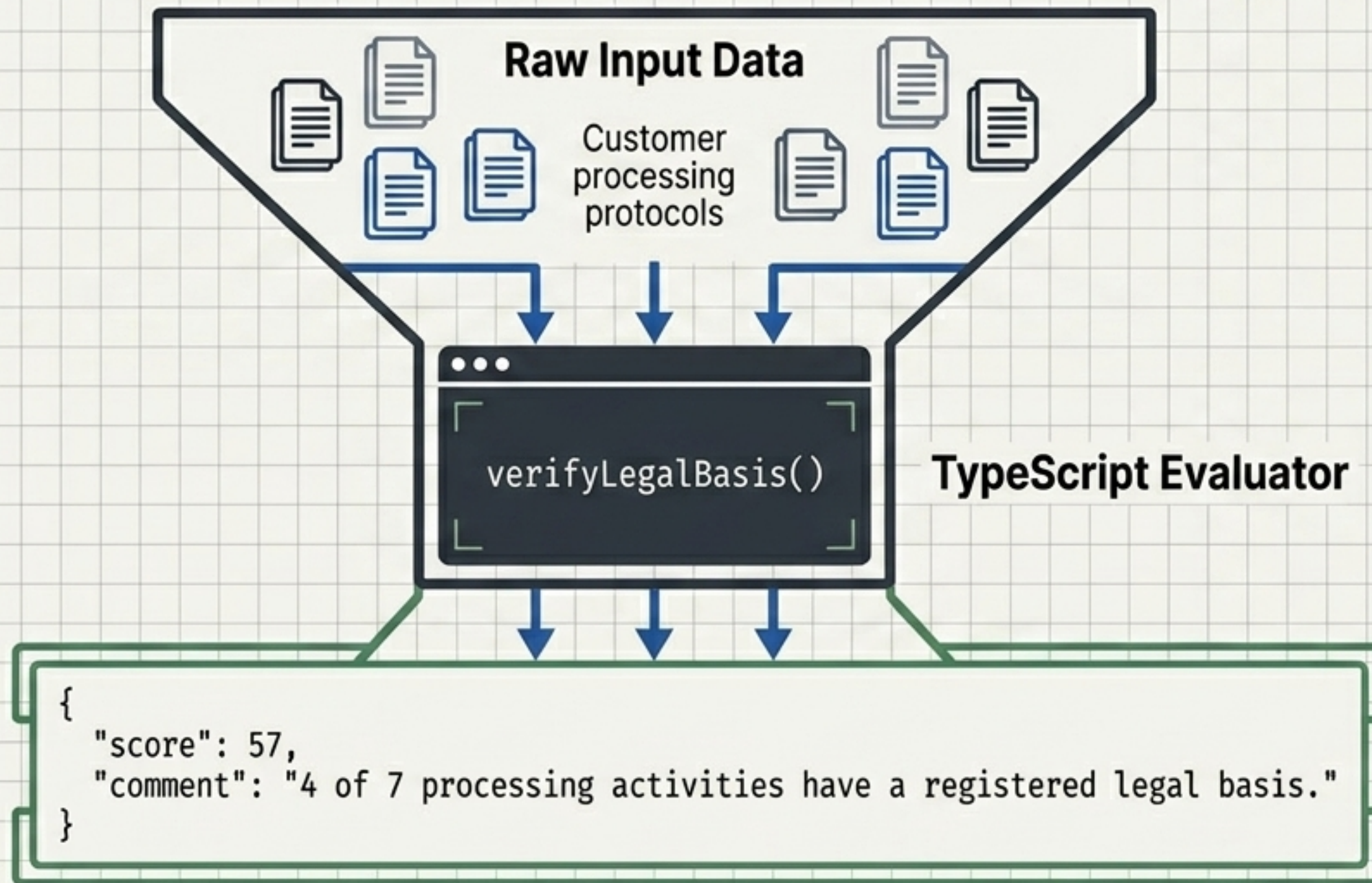


WIRING THE AGENT VIA THE KCP SIDECAR



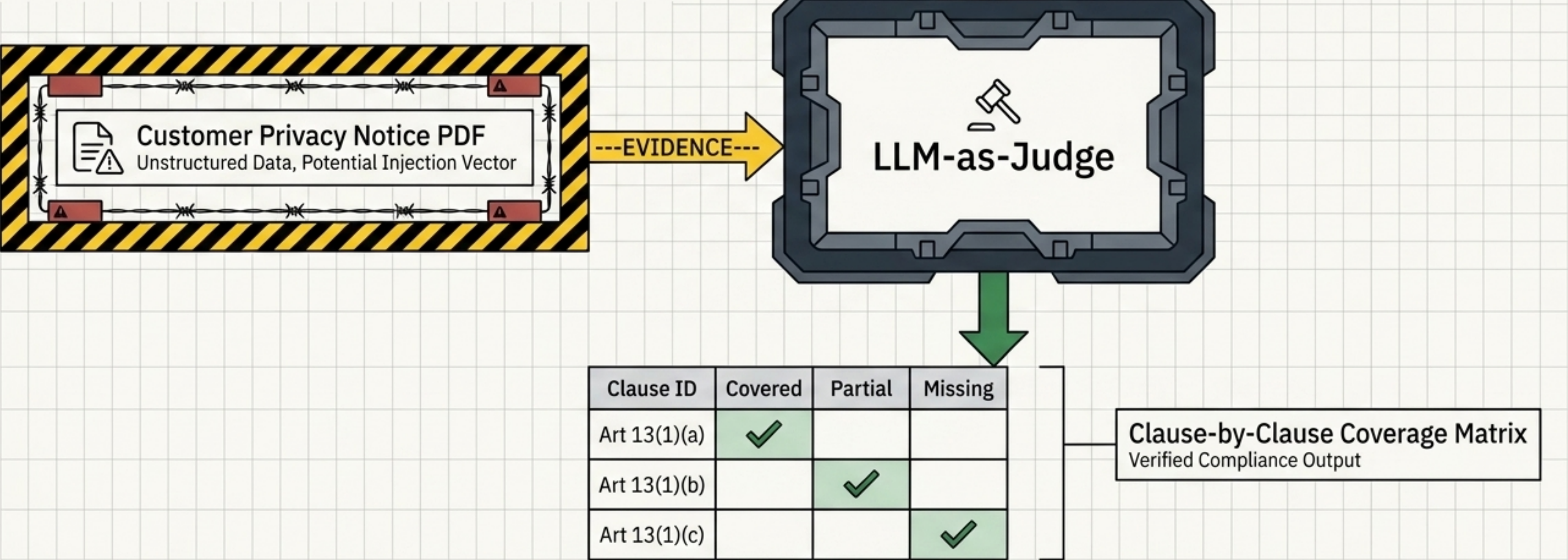
Evaluators measure real data, not AI vibes.

Move beyond chatbots. Over 178 TypeScript evaluator functions act as the ground truth across GDPR, NIS2, and ISO 27001. If the evaluator flags an error, the agent cannot claim compliance.



Fencing unstructured data for the LLM-as-Judge

Evaluating unstructured prose requires strict boundary controls. The system prompt explicitly defines customer PDFs as evidence, not instructions. Prompt injection defense is mandatory for compliance evaluation.



Encoding institutional legal judgment as versionable YAML

Expert lenses capture terminology corrections and adversarial review patterns, turning human interpretation into an automated, testable artifact applied to every subsequent review.

lens.yaml

terminologies:
corrections:

Before - Standard AI Translation

Ved vurdering av etterlevelse, må det dokumentere at de nødvendige prosessene er på plass. ⚠

Denne tilnærmingen kreves som regel for å sikre samsvar med gjeldende regelverk.

After - Expert Lens Applied

Ved vurdering av etterlevelse, må det påvise at de nødvendige prosessene er på plass. ✓

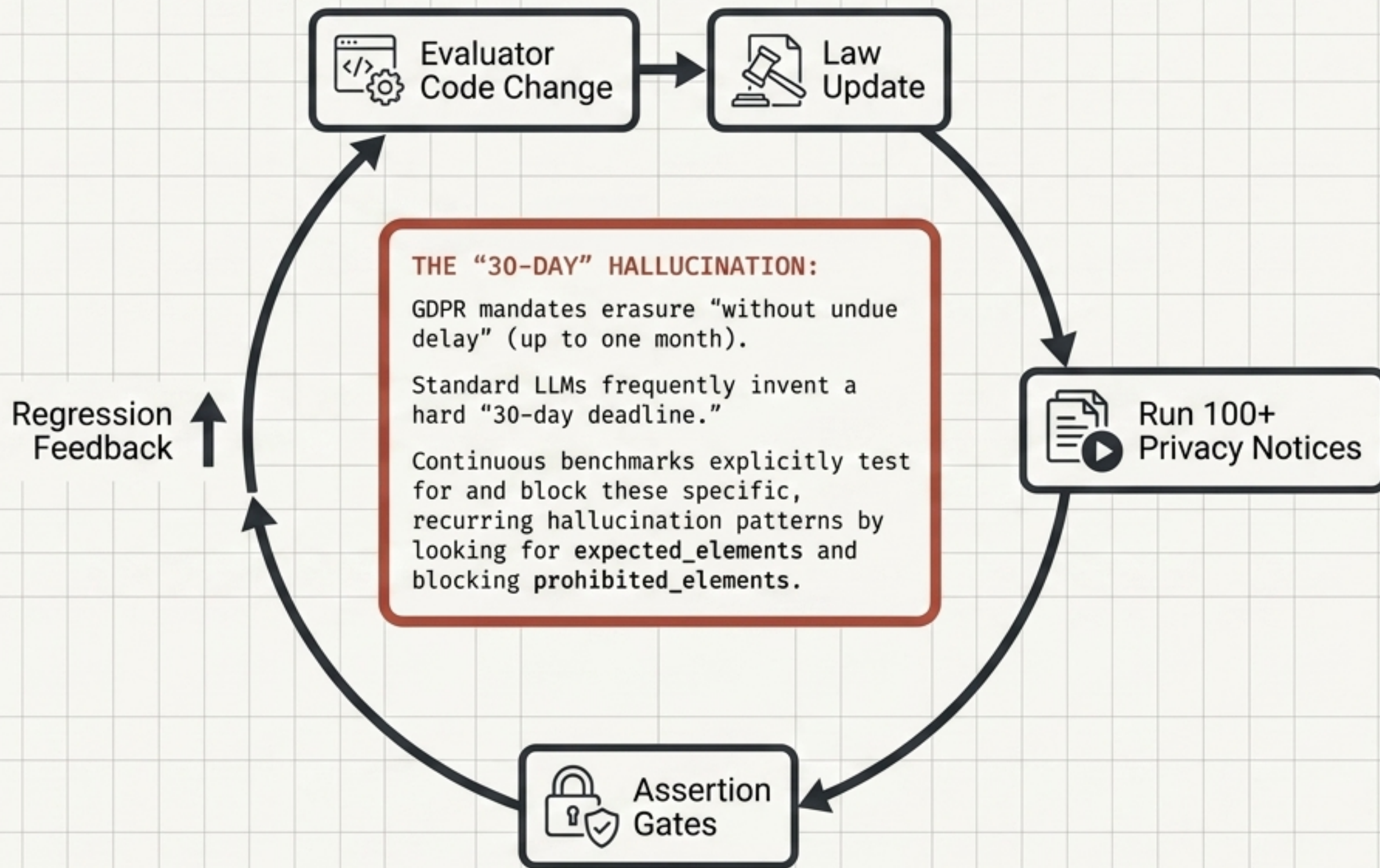
Denne tilnærmingen rettslig påkrevd for å sikre samsvar med gjeldende regelverk. ✓

← Legally distinct standard under Art 5(2)

← Legally required per Art 35(3)(b)

context:
- eioghore tsum orta
authority_references:
- Art 35(3)(b)

Continuous Benchmarking: CI/CD for AI.



Failure modes prevented by deterministic architecture



Without Authoritative Knowledge

Agent illegally approves employee health data consent (missing C-61/19 power imbalance rules). ⚠️



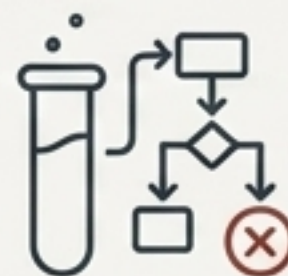
Without Evaluators

Agent approves compliance simply because a privacy notice exists, ignoring missing legal bases or DPOs. ⚠️



Without Signatures

System vulnerable to tampered regulatory text (e.g., a fabricated Article 6(1)(g) legal basis). ⚠️

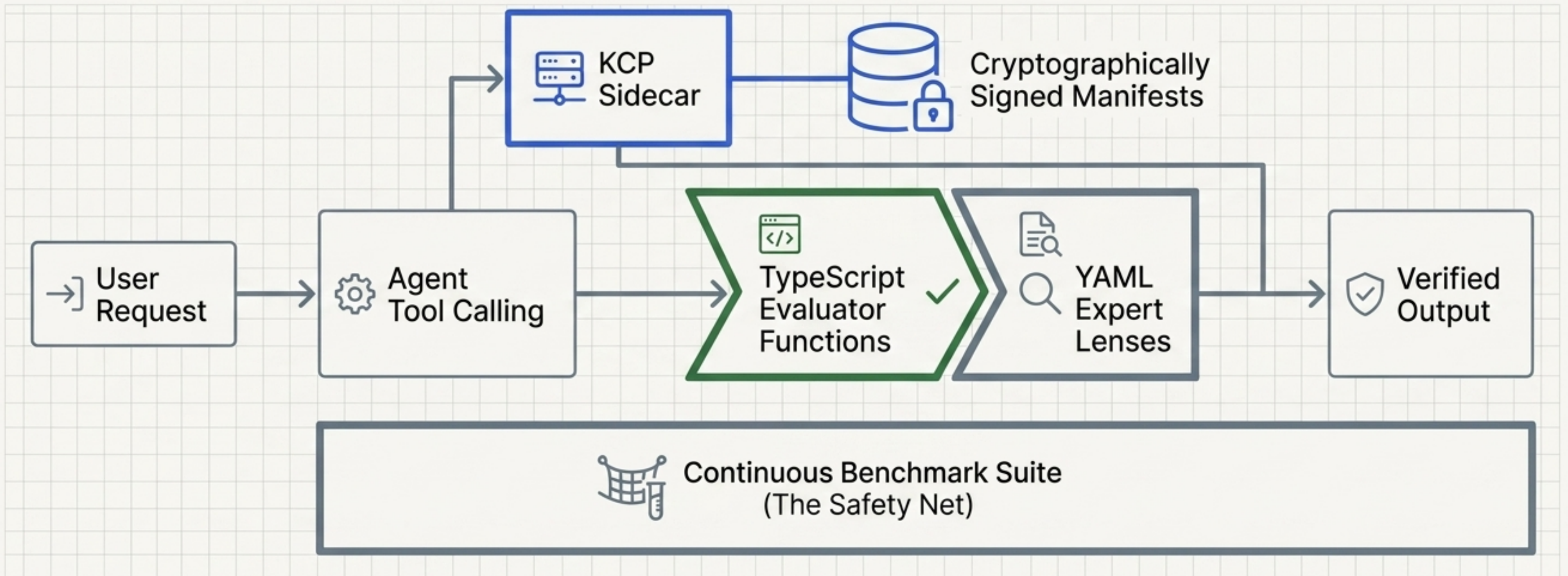


Without Benchmarks

Changes to the Art. 35 evaluator accidentally approve invalid DPIAs without triggering regression alerts. ⚠️

The Complete Production Architecture

Fragment extraction builds the knowledge. KCP manifests secure it. The sidecar serves it. Evaluators score it. Lenses refine it. Benchmarks protect it.



The 1-5-1-1 Implementation Framework

You do not need 178 evaluators to begin. Build these four components. Watch the agent's performance move from "plausible" to "provably correct." Then expand the footprint.

- 1 Regulation** (e.g., GDPR)
- 5 High-value Article Fragments** (Art. 5, 6, 9, 28, 32)
- 1 Evaluator Function** (e.g., Legal Basis checking)
- 1 Benchmark Test** (to catch the most common hallucination)

Your agent is only as good as what it knows.

Give it something authoritative to know. Then build the deterministic systems that prove it knows it correctly.



github.com/cantara/knowledge-context-protocol (Apache 2.0)

Derived from production architecture at Mynder.